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Euclidean Geometry In Mathematical Olympiads

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Problem 1.37 (BAMO 1999/2)



Angle Chasing

Chapter 1

fgm

fgmcon



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Mar 8, 2018, 9:23 PM • 3

#1

Problem 1.37 (BAMO 1999/2)

Let C be a circle in the xy -plane with center on the y -axis and passing through $A = (0, a)$ and $B = (0, b)$ with $0 < a < b$. Let P be any other point on the circle, let Q be the intersection of the line through P and A with the x -axis, and let $O = (0, 0)$. Prove that $\angle BQP = \angle BOP$

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#2

Coordinate Solution



The equation for the circle Γ is $x^2 + (y - \frac{a+b}{2})^2 = (\frac{b-a}{2})^2$. Observe that $\angle BPQ = 90^\circ$ by Thale's theorem. It follows that the circumcenter w of $\triangle BPQ$ lies on some point $(k, b/2)$. Through some algebra, we realize that $k = \frac{\sqrt{b^2 - 2ab}}{2}$. Hence, $w(\frac{\sqrt{b^2 - 2ab}}{2}, \frac{b}{2})$. Through the distance formula, observe that the radius of w is $\sqrt{\frac{b^2}{2} - \frac{ab}{2}} = r$. We now show that $wO = r$. This can be done by the distance formula, which indeed shows that $wO = \sqrt{\frac{b^2}{2} - \frac{ab}{2}} = r$. Thus, $\triangle BQP$ and $\triangle BOP$ share the same circumcircle, so B, Q, O, P are concyclic. The result follows. ■

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Mar 8, 2018, 9:34 PM • 3

#3

Man ■ is awesome.

Drunken_Master

327 posts

Mar 9, 2018, 3:05 AM • 9

#4

$\angle BOQ = \angle BPQ = 90^\circ$, so $BPOQ$ is cyclic. Hence, done. $\square$

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Mar 9, 2018, 6:40 AM • 5

#5

(Facepalm) why didn't I realize that.

Kuroshio

72 posts

Mar 24, 2020, 3:18 AM

#6

First note that AB is the diameter of the circle.

$$\angle BPQ = \angle BPA = 90^\circ = \angle BOQ$$

So points B, P, O, Q are concyclic. The conclusion follows.

This post has been edited 2 times. Last edited by Kuroshio, Jul 14, 2020, 2:19 AM

chi.wawa

31 posts

Mar 26, 2020, 5:12 PM

#7

Why can you come to the conclusion that because $BOQ=BPQ$, $BPOQ$ is concyclic?

Kagebaka

3001 posts

Mar 28, 2020, 7:43 PM

#8



chi.wawa wrote:

Why can you come to the conclusion that because $BOQ=BPQ$, $BPOQ$ is concyclic?

As others have already pointed out, $\angle BOP$ is equal to 90° since the x and y axes are perpendicular to each other, and $\angle BPQ = \angle BPA$ is equal to 90° since BA is a diameter of the circle.

bever209

1522 posts

Apr 13, 2020, 12:46 PM

#9



chi.wawa wrote:

Why can you come to the conclusion that because $BOQ=BPQ$, $BPOQ$ is concylic?

u can also use (iii) in lemma 1.9.

Archimedes15

1490 posts

Jun 24, 2020, 6:41 PM • 1 👍

#10

One-liner

This post has been edited 2 times. Last edited by Archimedes15, Jun 24, 2020, 6:41 PM